EE5239 Introduction to Nonlinear Optimization

Lecture 6b: Optimization Over Constraints (II)

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Summary

- When we are no longer "free" to explore the entire space, many interesting things happen

  1. Optimality condition needs to be updated
  2. Algorithm needs to be updated
  3. But surprisingly, most theory still works!

Summary

- After this lecture, I hope you will be able to understand the steps and rational behind
  1. Feasible direction method
  2. gradient projection (GP) method
  3. conditional gradient method
- Know how to implement the GP method for simple problems (e.g., quadratic minimization over box constrains)
- **Readings**
  1. Section 2.2-2.3 (version 2)
  2. Section 3.2-3.3 (version 3)

Announcement

- HW 3 will only grade the report; the codes will not be graded, but you still need to be careful about it because your subsequent HWs will be built upon those; [don't need to show classification results yet]
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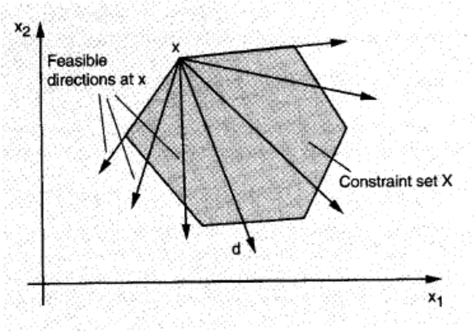
- HW 2 has been graded
- We will make comments to all the proposals submitted

Today's topic

- Today we will focus on discussing some general optimization methods to deal with constraints
- **General rule**: (1) walk towards descent directions, and (2) always stay feasible
- We will pay special attention to connection with GD-type methods

Feasible Directions

- A feasible direction at an $x \in X$ is a vector $d \neq 0$ such that $x + \alpha d$ is feasible for all sufficiently small $\alpha > 0$
- The set of feasible directions at x is the set of all $(z - x)$ where $z \in X$, $z \neq x$ (for convex sets).



Feasible Directions Method

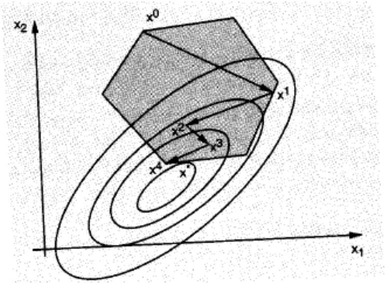
- A feasible direction method:

$$x^{r+1} = x^r + \alpha_r d^r,$$

where d^r is:

1. A feasible descent direction, i.e., $\nabla f(x^r)' d^r < 0$;

2. $\alpha_r > 0$ is such that $x^{r+1} \in X$.



Feasible Directions Method

- Alternative definition (for convex sets):

$$x^{r+1} = x^r + \alpha_r(\overline{x}^r - x^r) \tag{0.1}$$

where $\alpha_r \in (0, 1]$, $\bar{x}^r$ is some feasible point.

- If x^r is nonstationary,

$$\nabla f(x^r)'(\overline{x}^r - x^r) < 0. \tag{0.2}$$

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- **Stepsize rules**: Three common choices:
  1. Limited minimization:

$$\alpha_r = \arg \min_{\alpha \in [0,1]} f(x + \alpha(\bar{x} - x)) \tag{0.3}$$

  2. Armijo: $\alpha_r = \beta^{m_r} s$, where m_r is the first nonnegative m for which

$$f(x^r + s\beta^m(\overline{x}^r - x^r)) - f(x^r) \leq \sigma s \beta^m \nabla f(x^r)'(\overline{x}^r - x^r) \tag{0.4}$$

  3. Constant stepsize with $\alpha_r = 1$ for all r (a bit special)

Convergence Analysis

- Similar to the one for (unconstrained) gradient methods.
- The direction sequence $\{d^r\}$ is gradient related to $\{x^r\}$ if the following property can be shown: For any subsequence $\{x^r\}_{r\in K}$ that converges to a nonstationary point, the corresponding subsequence $\{d^r\}_{r\in K}$ is bounded and satisfies

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$$\limsup_{r\to\infty, r\in K} \nabla f(x^r)' d^r < 0.$$
- **Proposition (Stationary of Limit Points)** Let $\{x^r\}$ be a sequence generated by the feasible direction method $x^{r+1} = x^r + \alpha_r d^r,$. Assume that:
  $\star$ $\{d^r\}$ is gradient related
  $\star$ α_r is chosen by the limited minimization rule or the Armijo rule. Then every limit point of $\{x^r\}$ is a stationary point.
- Proof is nearly identical to the unconstrained case.

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$$x^{r+1} = x^r + \alpha_r(\overline{x}^r - x^r),$$
$$\bar{x}^r = \mathrm{proj}_X[x^r - s_r \nabla f(x^r)]$$

where, $\mathrm{proj}_X[\cdot]$ denotes projection on the set X, $\alpha_r \in (0,1]$ is a stepsize, and s_r is a positive scalar.

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- **Question:** why $(\overline{x}^r - x^r)$ is a good direction?
- **Hint:** the optimality condition for the projection?

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- When we refer to "GP" method, usually variant II; **Question:** connection between Variant II and GD?

Example: Nonnegative LS

- Consider the following nonnegative least square problem

$$\min \ \frac{1}{2}\|Ax - b\|^2, \quad \text{s.t.. } x \geq 0$$

- Lots of practical applications, especially useful when dealing with nonnegative data

Example: Nonnegative LS

- **Example**: nonnegative data such as pixel intensity values of an image, time measurements, histograms or count data, economical quantities such as prices, incomes and growth rates
- To build a meaningful linear model for these data requires that, the outcome is a nonnegative linear combination of the features
- Gradient projection?

$$x^{r+1} = \mathsf{proj}_{x\geq 0}\left[\underbrace{x^r - s_r(A^T(Ax^r - b))}_{\text{denote as } y}\right]$$

Perform the Projection

- Solve

$$\min \frac{1}{2}\|x - y\|^2, \quad \text{s.t.. } x \geq 0$$

- **Solution** [graphically]

$$x_i^* = y_i, \;\; \text{if } y_i \geq 0, \quad x_i^* = 0 \text{ otherwise}, \quad \forall\, i$$

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- **Why**? Check optimality condition

$$\sum_{i=1}^{K} \langle x_i^* - y_i, x_i - x_i^* \rangle \geq 0, \; \forall\, x \geq 0$$

- If $y_i \geq 0$, then $\langle x_i^* - y_i, x_i - x_i \rangle = 0$
- if $y_i \leq 0$, then $\langle 0 - y_i, x_i - 0 \rangle \geq 0$
- Verified that the [solution] satisfies the optimality condition

Convergence of GP Methods (Version I)

- Fix s, if α_r is chosen by the limited minimization rule or Armijo rule, every limit point of $\{x^r\}$ is stationary; [Prop. 2.3.1 in book]

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$$\nabla f(x^r)'(\overline{x}^r - x^r) < 0.$$

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- Note that $\{\overline{x}^r - x^r\}_{r\in K}$ is given by $\text{proj}_X[x^r - s\nabla f(x^r)] - x^r$. Using properties of projection

$$(x^r - s\nabla f(x^r) - \overline{x}^r)'(x - \overline{x}^r) \leq 0, \text{ for all } x \in X.$$

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- Applying this relation with $x = x^r$

$$\nabla f(x^r)'(\overline{x}^r - x^r) \leq -\frac{1}{s} \| x^r - \text{proj}_X[x^r - s\nabla f(x^r)] \|^2 < 0$$

Convergence of GP Methods (Version II)

- Similar conclusion for constant stepsize $\alpha_r = 1, s_r = s$; under a Lipschitz condition on ∇f, and for small enough stepsize
$$0 \leq s \leq \frac{2}{L}.$$

- See Prop. 2.3.2 in book (V2)

- Similar conclusion for Armijo rule for this case.

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- So we have

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&\leq \| (x^r - x^*) - s(\nabla f(x^r) - \nabla f(x^*)) \| \\
&= \| (I - sA)(x^r - x^*) \| \\
&\leq \|I - sA\| \|(x^r - x^*)\| \\
&\leq \max\{|1 - s\lambda_{\min}|, |1 - s\lambda_{\max}|\} \| x^r - x^* \| \\
&\leq \left(\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}}\right) \| x^r - x^* \| = \left(1 - \frac{2}{\kappa + 1}\right) \| x^r - x^* \|
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In the last inequality we choose

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Convergence Rate Analysis

- Generally speaking, for strongly convex problems
  1. Convergence rate depends on $\kappa = \lambda_{\max}/\lambda_{\min}$; that is, the condition number!
  2. This is similar to the unconstrained cases
  3. Requires about $\kappa \ln(1/\epsilon)$ to find ϵ-relative optimal solution

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- Slow (sublinear) convergence, even when the problem is strongly convex? Then what's the benefit??

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- Assume that X is compact, so $\bar{x}^r$ is guaranteed to exist.
- Slow (sublinear) convergence, even when the problem is strongly convex? Then what's the benefit??
- **Subproblem easy, no projection needed!!** [Example: the simplex constraints]

Convergence of the Conditional Gradient Method

- Show that the direction sequence of the conditional gradient method is gradient related, so the generic convergence result applies.

- Suppose that $\{x^r\}_{r \in K}$ converges to a nonstationary point $\tilde{x}$. We must prove that

$$\{\| \overline{x}^r - x^r \|\}_{r \in K}: \text{ bounded}, \limsup_{r \to \infty, r \in K} \nabla f(x^r)'(\overline{x}^r - x^r) < 0.$$

- 1st relation: Holds because $\overline{x}^r \in X$, $x^r \in X$, and X is compact.

- 2nd relation: Note that by definition of $\overline{x}^r$,

$$\nabla f(x^r)'(\overline{x}^r - x^r) \leq \nabla f(x^r)'(x - x^r), \text{ for all } x \in X$$

Convergence of Conditional Gradient

Taking limit as $r \to \infty, r \in K,$ minimizing the RHS over $x \in X$, and using the nonstationary of $\widetilde{x}$,

$$\limsup_{r \to \infty, r \in K} \nabla f(x^r)'(\overline{x}^r - x^r) \leq \min_{x \in X} \nabla f(\widetilde{x})'(x - \widetilde{x}) < 0,$$

thereby proving the 2nd relation.

- For stepsize $\alpha_r = 2/(2 + r)$ the convergence rate is $\mathcal{O}(1/r)$ [On board] [Exercise 2.2.3, 2.2.4]